A Federated Multi-Task Learning Approach using Small Language Models for Natural Language Understanding Tasks

1. INTRODUCTION

- Motivation: Privacy-preserving and collaborative learning for NLU.
- Limitations of centralized and single-task NLU models.
- Challenges of federated learning with heterogeneous NLU tasks and non-IID data.
- Advantages of Small Language Models in federated settings (communication efficiency, deployability).
- Research objectives and contributions.

2. RELATED WORK

2.1 Natural Language Understanding Tasks

- Overview of core NLU tasks (text classification, NER, sentiment analysis, NLI, intent detection).

2.2 Small Language Models

- DistilBERT, MiniLM, ALBERT, efficiency-focused architectures.

2.3 Multi-Task Learning in NLP

- Hard vs soft parameter sharing.
- Task balancing and negative transfer.

2.4 Federated Learning for NLP

- Federated algorithms (FedAvg, FedProx, FedMultiTask).
- Data heterogeneity and communication challenges.

2.5 Summary and Research Gap

3. PROBLEM DEFINITION

- Formalization of federated multi-task NLU.
- Definition of clients, tasks, datasets, and privacy constraints.
- Optimization objective under FL and MTL.

4. PROPOSED METHOD

4.1 System Overview

- Federated learning architecture with multi-task objectives.
- Client-server communication workflow.

4.2 Small Language Model Backbone

- Selection of SLM architectures.
- Shared encoder and task-specific output heads.

4.3 Multi-Task Learning Strategy

- Parameter sharing across tasks.
- Task-specific losses and balancing mechanisms.

4.4 Federated Optimization

- Aggregation strategy (FedAvg / FedProx / task-aware aggregation).
- Handling non-IID data and client heterogeneity.

4.5 Communication-Efficient Fine-Tuning

- Parameter-efficient fine-tuning (LoRA, Adapters).
- Communication cost reduction strategies.

5. EXPERIMENTS

5.1 Experimental Setup

- Datasets and Tasks

- NLU benchmarks (SST-2 for sentiment analysis task, QQP for paraphrase detection task, STS-B semantic similarity task).
- Federated data partitioning strategies (IID vs non-IID).

- Baselines

- Centralized multi-task training.
- Federated single-task learning.
- Large model-based federated approaches.

- Evaluation Metrics

- Task-level metrics:
  - SST-2: sentiment analysis task: accuracy & F1
  - QQP: paraphrase detection task: accuracy & F1
  - STS-B: semantic similarity task: Pearson correlation & Spearman correlation.
- Federated metrics (communication rounds, convergence speed).

- Implementation Details

- Training configuration, hardware, hyperparameters.

5.2 Results

- Overall Multi-Task Performance

- Performance comparison across tasks and baselines.

- Impact of Federated Learning

- Effect of data heterogeneity and client variability.

- Efficiency Analysis

- Communication cost, training time, model size.

- Ablation Studies

- Effect of SLM choice, task sharing strategy, and fine-tuning method.

5.3 Discussion

- Trade-offs between privacy, performance, and efficiency.
- Insights into task transfer and negative transfer.
- Practical deployment considerations.

6. CONCLUSION

- Summary of contributions and findings.
- Implications for scalable and privacy-preserving NLU systems.
- Support for more diverse NLU tasks and languages.
- Advanced federated optimization and personalization.
- Continual and cross-domain federated learning.

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